Baseline Verification — Synthesis Report

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4DVarNet-FM: Baseline Verification Report
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Common parameters:
  T_max = 3.0s    dt = 0.01    Steps = 300
  obs_interval = 20    R_var = 0.5    B_var = 2.0

DA Baseline parameters:
  DA window steps (DWS) = 50
  Weak-4DVar : opt_steps=150, lr=0.02
  Strong-4DVar: max_iter=40, lr=0.1
  EnKF       : N_ensemble=30, inflation=1.2

Case studies:
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Parameter                    CS1                                    CS2
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  Forcing                    Noise-free (true W_L)                  Corrupted (OU process)
  Parameter bias             0%                                     15%
  Forcing-state bias         0.0                                    0.15
  Coupling                   Linear                                 Quartic
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Test data:
  200 trajectories per case study, generated with seed=123 (CS1) / 124 (CS2)
  Forcing corrupted by Ornstein-Uhlenbeck noise (tau_eta=5.0, sigma_eta=0.707)
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Baseline Verification — Metrics

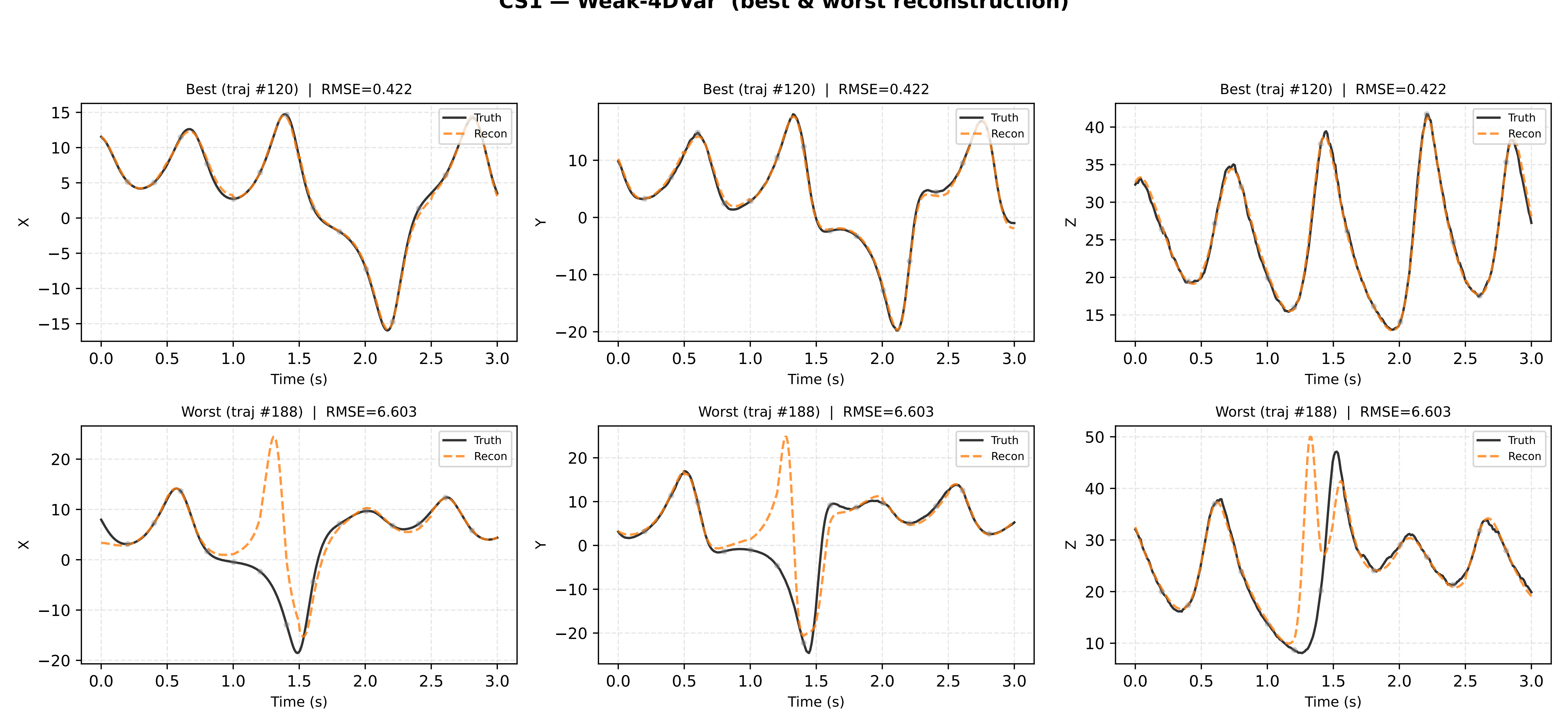
Summary of Results – Mean RMSE

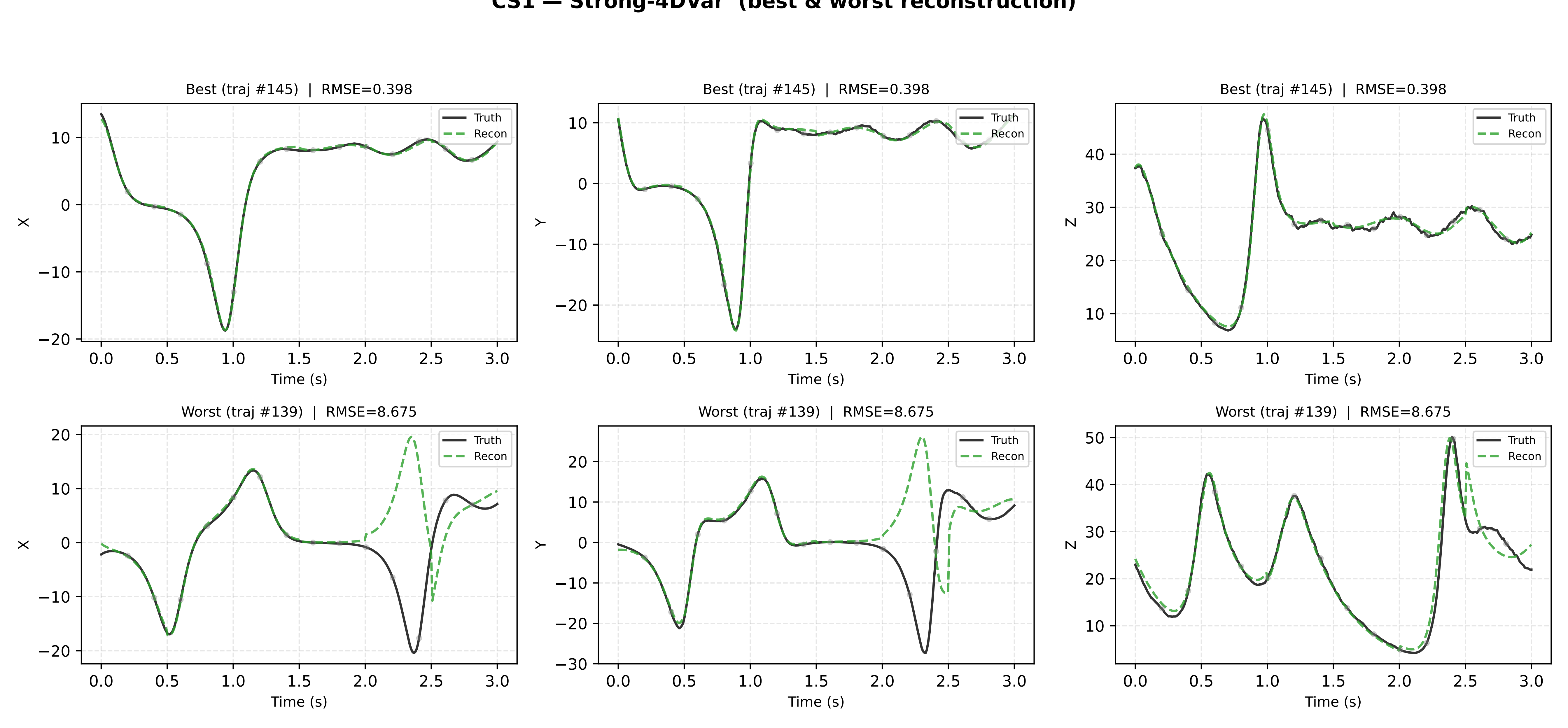
Method	CS1 X	CS1 Y	CS1 Z	CS1 μ	CS2 X	CS2 Y	CS2 Z	CS2 μ	Deg	
Weak-4DVar	0.4888	0.6807	0.7933	0.6543	0.7228	1.1954	2.7309	1.5497	2.37	x
Strong-4DVar	0.4889	0.6850	0.8074	0.6604	0.9620	1.4219	3.6121	1.9987	3.03	x
EnKF	0.5398	0.8538	0.9534	0.7823	1.3188	2.0582	4.3544	2.5772	3.29	x

Best in CS1 μ: Weak-4DVar (0.6543)
Best in CS2 μ: Weak-4DVar (1.5497)

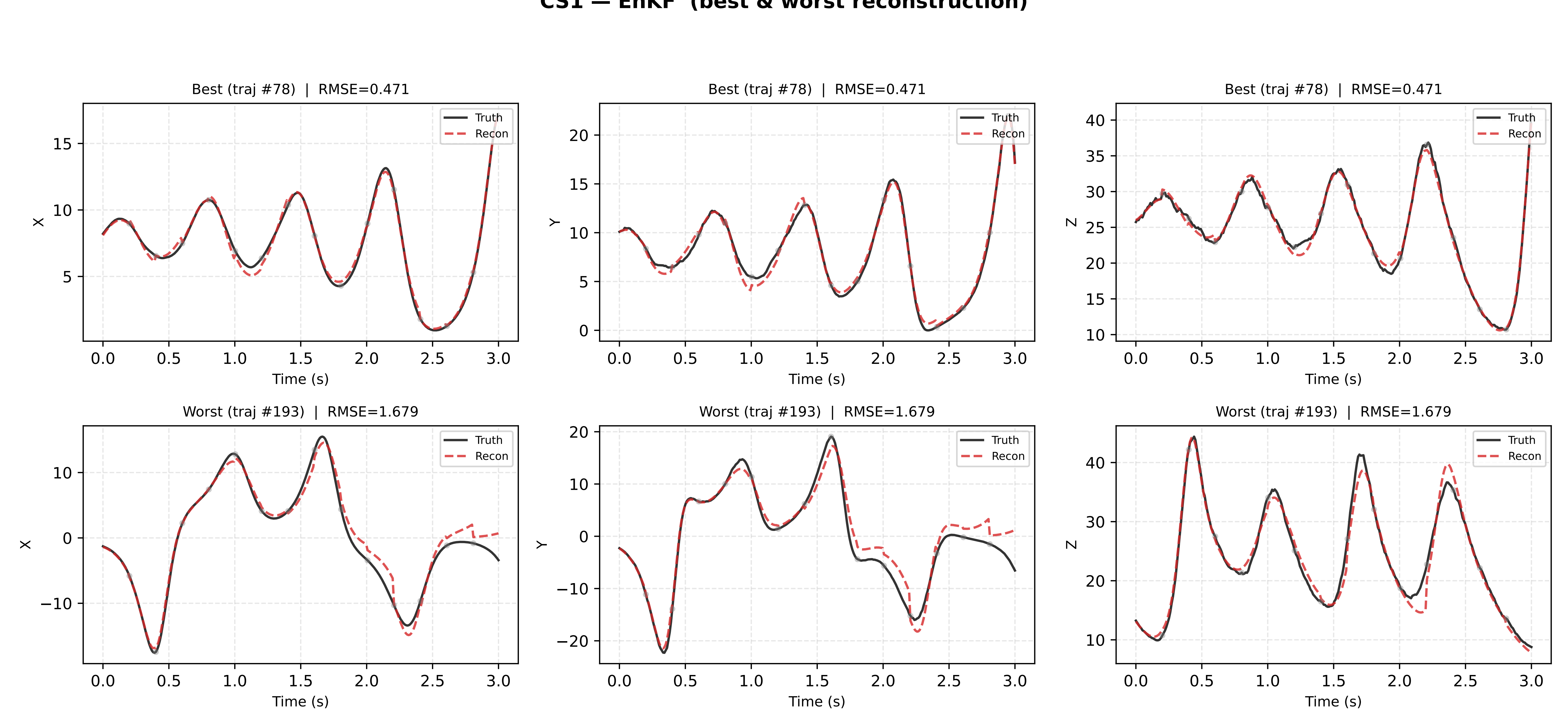
Degradation = CS2 μ / CS1 μ.   Lower = more robust.

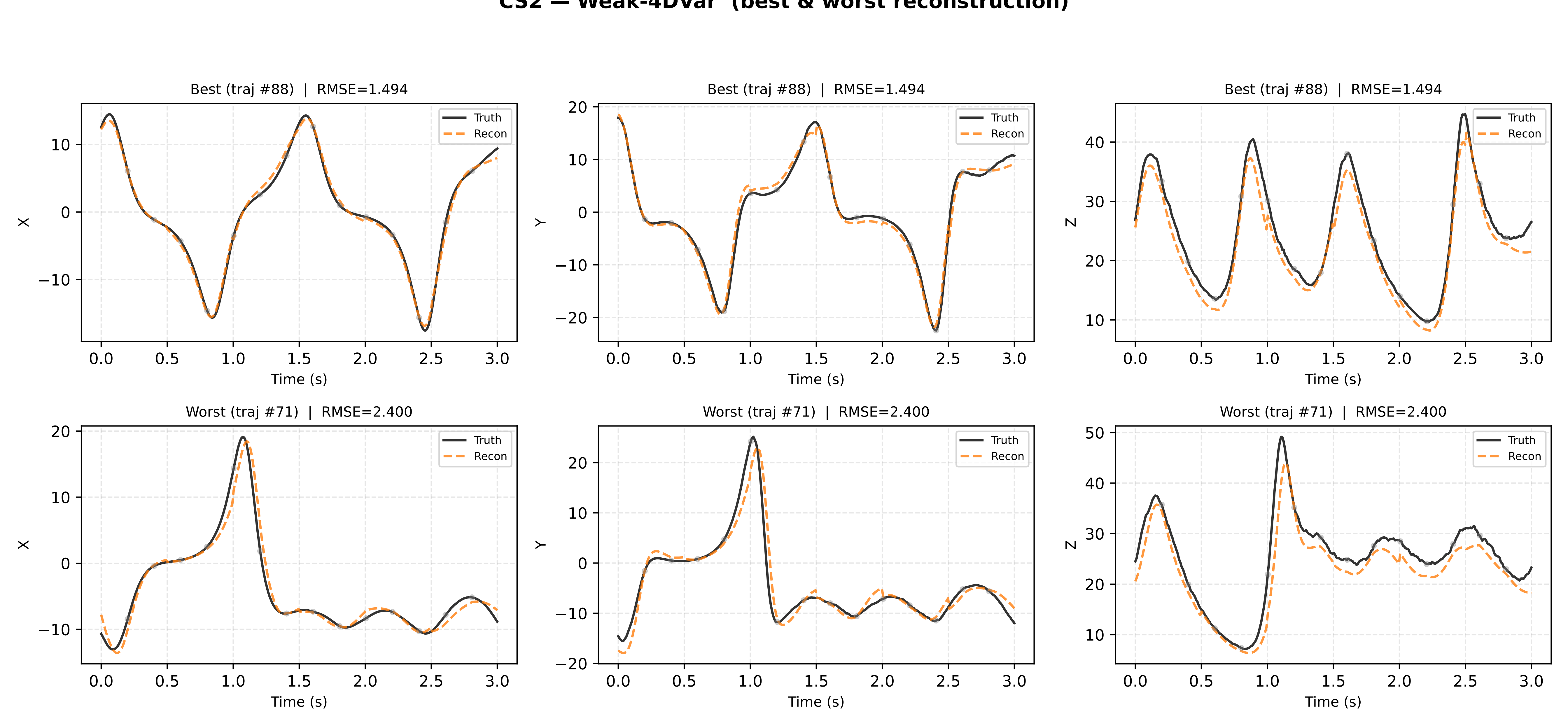
Config file: experiments/baselines_dws50_inf1.2.json


CS1 — Weak-4DVar (best & worst reconstruction)
Best (traj #120) | RMSE=0.422
Worst (traj #188) | RMSE=6.603
Truth
Recon
X
Y
Z
Time (s)



CS1 — Strong-4DVar (best & worst reconstruction)
Best (traj #145) | RMSE=0.398
Worst (traj #139) | RMSE=8.675
Truth
Recon
X
Y
Z
Time (s)


CS1 — EnKF (best & worst reconstruction)
Best (traj #78) | RMSE=0.471
Worst (traj #193) | RMSE=1.679
Truth
Recon
X
Y
Z
Time (s)


CS2 — Weak-4DVar (best & worst reconstruction)
Best (traj #88) | RMSE=1.494
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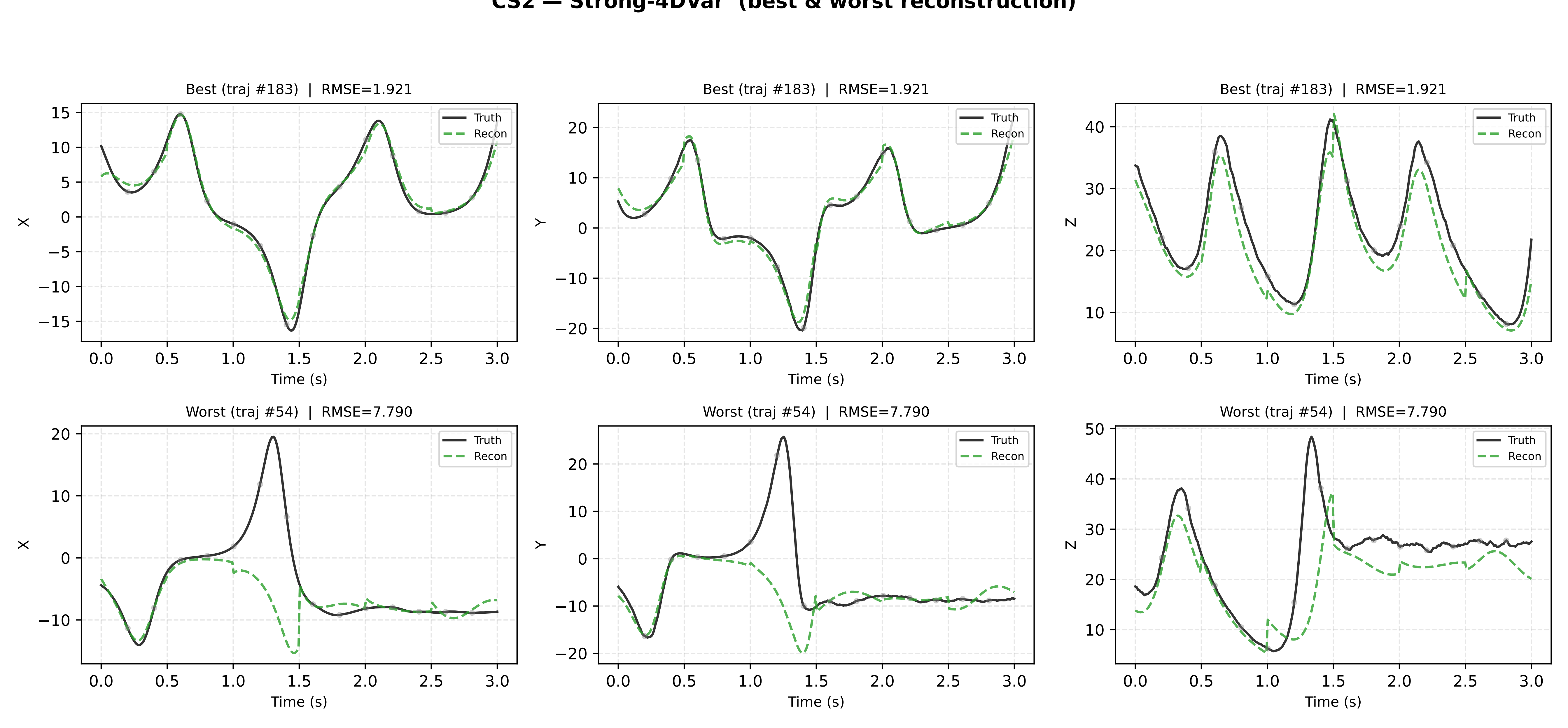
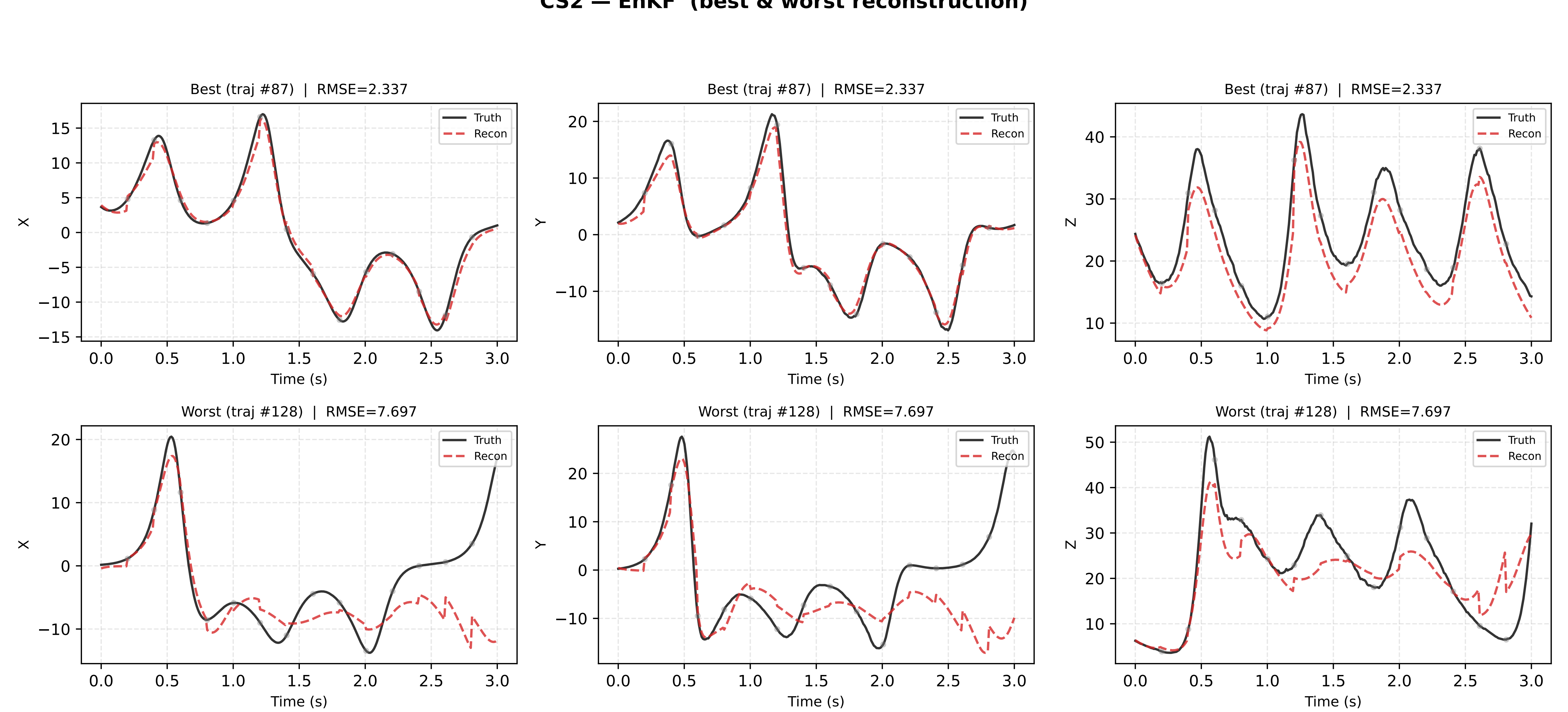
Worst (traj #71) | RMSE=2.400
Worst (traj #71) | RMSE=2.400
Worst (traj #71) | RMSE=2.400
Truth
Recon
X
Y
Z
Time (s)



CS2 — Strong-4DVar (best & worst reconstruction)
Best (traj #183) | RMSE=1.921
Worst (traj #54) | RMSE=7.790
Truth
Recon
X
Y
Z
Time (s)


CS2 — EnKF (best & worst reconstruction)
Best (traj #87) | RMSE=2.337
Worst (traj #128) | RMSE=7.697
Truth
Recon
X
Y
Z
Time (s)